Arista Access Node JAY-ACC-47 SOP for Optical Transport Failure

Node: JAY-ACC-47 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9315

Description

The Arista Access Node JAY-ACC-47 at the Jayanagar Metro Site is equipped with Arista's EOS (Extensible Operating System) running on a 7250 CX3 platform. This platform is used for high-capacity aggregation and edge routing. The failure domain of interest, Optical Transport Failure, can occur due to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. This is a Major failure event (Alarm Code: ALARM_9315) because it directly impacts the network's ability to transport data, affecting both the control and data planes. The underlying protocols and hardware mechanics involved in this failure domain are primarily related to the Optical Transport Network (OTN) and the SFP/XFP interfaces. The OTN is responsible for the optical transmission of data between nodes, while the SFP/XFP interfaces provide the physical connection to the fiber optic cables. When an SFP/XFP degrades, the received optical signal may be weakened, leading to errors and eventually a failure. Similarly, fiber cuts or Rx/Tx power loss can disrupt the optical signal, causing a failure. This SOP aims to troubleshoot and resolve such failures on the Arista Access Node JAY-ACC-47.

Pre-requisite Checks

- Pre-requisite: Remote hands to be dispatched to the Jayanagar Metro Site for physical access and assistance as required., Description: This is necessary to physically inspect the node, check the SFP/XFP interfaces, and perform any necessary actions.

- Pre-requisite: Redundant link status to be checked and verified., Description: This is essential to determine if the failure is isolated to a single link or if it affects the entire network.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface Status: * SFP-GE1/0/1: connected, SFP type: 100GBase-LR, Description: This command checks the status of the interfaces, including the SFP/XFP status, which can indicate if there is a problem with the physical connection.

Command: show system internal optics, Expected Output: Optics Status: * SFP-GE1/0/1: optics status: OK, Temperature: 25°C, Description: This
```

```
command provides detailed information about the optics, including
temperature and status, which can help identify issues with the
SFP/XFP.

Command: show system internal fiber, Expected Output: Fiber Status: *
Fiber 1: fiber status: OK, Length: 10km, Description: This command
checks the status of the fiber optic cables, including length and
status, which can help determine if there is a problem with the fiber.
```

Resolution Steps

Step 1: Step: Power down the node, Command: power shut-down node, Description: This step is necessary to ensure the node is powered down before performing any physical repairs or replacements.

Step 2: Step: Replace the SFP/XFP, Command: replace sfp ge1/0/1, Description: This step involves replacing the SFP/XFP module with a new one. The new module should be of the same type (e.g., 100GBase-LR) and should be installed according to the manufacturer's instructions.

Step 3: Step: Verify the fiber connection, Command: verify fiber connection, Description: This step involves checking the fiber optic cables to ensure they are properly connected and not damaged.

Post-Check Verification

```
Command: show interface status, Expected Output: Interface Status: *
SFP-GE1/0/1: connected, SFP type: 100GBase-LR, Description: This
command checks the status of the interfaces, including the SFP/XFP
status, which should indicate that the interface is now connected and
functioning properly.

Command: show system internal optics, Expected Output: Optics Status:
* SFP-GE1/0/1: optics status: OK, Temperature: 25°C, Description: This
command provides detailed information about the optics, including
temperature and status, which should indicate that the optics are now
functioning properly.
```

Roll-back Procedures

- Step: Power up the node, Command: power start node, Description: This step is necessary to power up the node in case the fix fails.

- Step: Restore the original SFP/XFP, Command: replace sfp ge1/0/1, Description: This step involves restoring the original SFP/XFP module. The original module should be of the same type (e.g., 100GBase-LR) and should be installed according to the manufacturer's instructions.